



NATIONAL SCHOOL OF BUSINESS MANAGEMENT

ONLINE EXAMINATION-2020 JUNE

BSc (Hons) Software Engineering/ BSc Management Information Systems (Special)-16.2/17.1

4th Year 1st Semester Examination

20-June-2020

CS407.3 – Internet of Things

Instructions to Candidates

- 1) Answer ALL Questions**
- 2) This paper has two parts:**
 - **Part A: Multiple Choice Questions (25 Marks)**
 - **Part B: Structured (75 Marks)**
 - **Answer all the questions (both Part A and B) on the answer booklet.**
- 3) Total number of pages – Twelve (12)**
- 4) Use the provided answer booklet to write your answers and upload to the LMS as a PDF file within the stipulated time. This PDF file should be saved with your index no.**
- 5) You are required to submit the answer file within the allowed timeframe, and no additional time is provided for submissions.**
- 6) All the submissions will be checked for plagiarism. Plagiarism, collusion, and copying are serious offences in the university and serious penalties that would be imposed.**
- 7) You're not allowed to publish or disseminate any part of the paper online or offline.**

PART A: Multiple Choice Questions

(Total: 0.5 *50 =25 Marks)

1. Which protocol is used to link all the devices in the IoT?

- a) TCP/IP
- b) Network
- c) UDP
- d) HTTP

2. What is the bit size of the 8051 microcontroller?

- a) 8-bit
- b) 4-bit
- c) 16-bit
- d) 32-bit

3. What is the microcontroller used in Arduino UNO?

- a) ATmega328p
- b) ATmega2560
- c) ATmega32114
- d) AT91SAM3x8E

4. Which of the following protocols is used for WWW?

- a) FTP
- b) http
- c) w3
- d) All of the above

5. The first page that you normally view at a website is its:

- a) Home page
- b) Master page
- c) First page
- d) None of the above

6. Voice mail, E-mail, Online service, the Internet and the WWW are all example of

- a) Computer categories
- b) Connectivity
- c) Telecommuting
- d) None of the above

7. In reality, Internet protocol recognizes only

- a) An IP address
- b) A location of the host
- c) A postal mail address
- d) None of the above

8. -----is being used by internet of Things

- a) Radio identification technology
- b) Satellite
- c) Cable
- d) Broadband

9. Which is not one of the features of Internet of Things devices?

- a) Remotely controllable
- b) Programmable
- c) Can turn themselves off if necessary
- d) All are features

10. How many devices are estimated to be connected to the IoT by 2020

- a) Approximately 2 Billions
- b) Approximately 30 Billions
- c) Approximately 75 Billions
- d) Approximately 100 Billions

11. Which one is not an element of IoT

- a) People
- b) Process
- c) Security
- d) Things

12. The huge number of devices connected to the Internet of Things has to communicate automatically, not via humans. What is this called?

- a) Internet of Services
- b) Bot 2 Bot
- c) M 2 M
- d) Internet of content

13. WiFi uses how much frequency?

- a) 2.2GHz
- b) 3GHz
- c) 3.5GHz
- d) 2.4GHz

14. Who invented the internet or working protocol, to make communication possible?

- a) Mr Vint Cerf and Mr Bob Kahn
- b) Bill Gates
- c) Steve Jobs and Kevin Ashton
- d) Arthur Richter and Deborah W. Smyth

15. _____ are pre built circuit boards that fit on top of Android.

- a) Sensor
- b) Data types
- c) Breadboard
- d) Shields

16. How many digital pins are there on the UNO board?

- a) 14
- b) 12
- c) 16
- d) 20

17. A program written with the IDE for Arduino is called _____

- a) IDE source
- b) Sketch
- c) Cryptography
- d) Source code

18. Which sensor can detect nearby objects?

- a) Proximity sensor
- b) Humidity sensor
- c) Touch sensor
- d) Pressure sensor

19. The monitoring of machines, gears and objects are achieved by which sensor?

- a) Humidity sensor
- b) Proximity sensor
- c) Touch sensor
- d) Pressure sensor

20. _____ empowers IoT by bringing together everyday objects.

- a) Intelligence
- b) Connectivity
- c) Dynamic Nature
- d) Enormous Scale

21. Which is NOT one of the concerns of the Internet of Things?

- a) Data storage standards
- b) Efficiency
- c) Privacy concerns
- d) Cyber security

22. What is IoT

- a) Objects, things, sensors or devices are connected through communication interface
- b) Things connected through Wi-Fi to server
- c) Intelligent data processing and management through onside application
- d) None of the above

23. IoT hardware consisted with;

- a) Communication, Machine to Machine, Software (IDE), Cloud Platforms
- b) Communication, Protocols, Application, Cloud Platforms
- c) Communication, Protocols, Software (IDE), Cloud Platforms
- d) Communication, Protocols, Software (IDE), Sensors

24. A protocol is a standard for;

- a) A set of rules that allow electronic devices to communicate with each other
- b) Connecting device's in to an Arduino board
- c) Communicate sensors without wire connection
- d) Protocol is a programming language

25. Integrated development environment is

- a) Hardware application
- b) Software Application
- c) Communication application
- d) Cloud platforms – Cloud IoT

26. Binary Code consists with.

- a) 0s and 1s
- b) 0 and 1
- c) 10 and 01
- d) 0 to 9

27. How to write number “19” in Binary cord?

- a) 10001
- b) 10011
- c) 10001
- d) 11001

28. Calculate $0110 + 0111$ is;

- a) 31
- b) 21
- c) 13
- d) 12

29. ASCII is stand for;

- a) American Structure Code Information Interchange
- b) American Standard Code Information Interchange
- c) American Standard Code Information Interexchange
- d) Arduino Standard Code Information Interchange

30. How computer understand the 1 and 0

- a) Microprocessor switches ON and OFF
- b) Transistor switches ON and OFF
- c) Using inbuilt programmed circuits
- d) Using Microprocessor RAM

31. Microprocessor physically composed with;

- a) Control unit, Register, Arithmetic and Logic Units (ALU)
- b) Communication, Protocols, Software (IDE), Cloud Platforms
- c) Sensors, Connectivity, Application
- d) Processing unit and additional peripherals in the same unit (Chip)

32. Microprocessor physically composed with;

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33. ATmega328 specifications are

- a) Support the data up to eight (8) bits and 32KB internal built-in memory.
- b) 1KB Electrically Erasable Programmable Read Only Memory (EEPROM)
- c) Advanced RISC architecture, good performance, low power consumption, real timer counter having separate oscillator,
- d) All are features

34. Example for actuators

- a) Proximity sensor, Humidity sensor, Temperature sensor
- b) Lights, LED's, Motors, Speakers, Displays (LCD)
- c) Photoresistors, Thermistors, Ultrasound
- d) All are features

35. Radio Frequency Identification or RFID is technology work in

- a) GSM and GPRS
- b) Wi-Fi
- c) Radio frequency or radio waves
- d) Connect through internet

36. How Does RFID Work?

- a) Sensors, Connectivity and Gateways
- b) Sensor, Software (IED) and cloud storage
- c) Application Transceiver, Transponder and Antenna
- d) All are features

37. What are the two type of RFID

- a) Cloud and onsite
- b) Passive and Active
- c) With GSM connection and without GSM connection
- d) Data read directly and indirectly

38. What are the satellite services?

- a) Remote sensing and crop monitoring
- b) Internet access and navigation
- c) Communication, Radio and Television
- d) All are features

39. NAVSTAR - GPS owned by;

- a) Europe Union
- b) Russian Federation
- c) United States
- d) Japan

40. What are the segments in GPS consolation?

- a) Control segment, Space segment and User segment
- b) Control segment, monitoring segment and User segment
- c) Control segment and Space segment
- d) None of the above

41. Why Errors on GPS signal, due to

- a) Orbit, clock, Ionosphere and Multipath
- b) Speed of light and time difference
- c) Low power and Antenna problems
- d) When measuring Travel time

42. A “thing” has NOT a property

- a) Usually powered by battery
- b) Generally small in size
- c) It doesn't usually perform complicated tasks
- d) Providing link between two nodes

43. The two reasons to handover GSM connection

- a) No connections available in BTS
- b) Radio Signal's quality, power decreases and Traffic Capacity maximum
- c) To achieve Multiple access technology
- d) Time, Frequency and Code

44. Arduino programs can be divided in three main parts:

- a) Structure, Values (variables and constants), Functions
- b) Type, Name and Value
- c) Software, hardware and sensor
- d) Sketch, Programming and coding

45. Arduino IDE consists of 2 functions. What are they ?

- a) Build () and loop ()
- b) Setup () and build ()
- c) Setup () and loop ()
- d) Loop () and build () and setup ()

46. Arduino shields are also called as

- a) Extra peripherals
- b) Add on modules
- c) Connectivity modules
- d) Another arduinos

47. A function is a series of programming statements that can be called by name. Which command is called once when the program starts?

- a) loop()
- b) setup()
- c) (output)
- d) (input)

48. Which of the following variable types is NOT recognized by the Arduino

- a) float
- b) int
- c) long
- d) All are features

49. What is the range of numbers you can store in a variable of type **byte**?

- a) 0 to 9
- b) 0 to 65535
- c) -128 to 127
- d) 0 to 4294967295
- e) 0 to 255

50. To subtract 1 from a variable **counter**, you can use the following statement(s) (check all that apply)

- a) Counter - - ;
- b) Decrement (counter);
- c) Counter – 1;
- d) None of the above

PART B: Structured Essay

(Total: 75 Marks)

Question 1.

(Total: 25 Marks)

- a. What Is the Internet of Things (IoT) stands for? Explain with two examples
- b. In IoT development process, Wireless Communications affect the development and implementation of the Internet of Things (IoT)? Explain how it could be affected.
- c. What Are The Important Components Of the Internet Of Things?
- d. What Are the Main Challenges of the Internet of Things (IoT)?
- e. What is a “thing” In the context of the Internet of Things (IoT)?
- f. Explain two industries can use IoT platforms to improve their productivity?

Question 2.

(Total: 25 Marks)

- a. What is a Binary Code and how binary cord understanding numbers? Explain?
- b. What you will gain when you upgraded your system from 8bit and 16bit, explain?
- c. What is Microprocessor? and advantages/disadvantage of Microprocessor
- d. What is Microcontroller and Characteristics? Explain?
- e. Arduino is a platform using the sketch this platform work in the IoT environment, Explain
- f. Short notes;
 - a. How do computers understand binary cords?
 - b. What are active tags and passive tags in RFID
 - c. What are satellites?
 - d. Name 5 application used by GPS
 - e. What are Internet and Protocols?

Question 3.

(Total: 25 Marks)

You know that, Coronavirus disease 2019 (COVID-19) is an infectious disease caused by severe acute respiratory syndrome coronavirus. The viruses primarily spread between people during close contact, most often via small droplets produced by coughing, sneezing, and talking. In Sri Lank in

2nd May 2020, detect the first COVID 19 patients and now it has increased up to 1600 (+). As a software engineer, you have to designed application using Internet of Things (IoT) for Health Care Sector? The IoT has many applications in the health care field, in both health monitoring, treatment and including tracking. You have to designed concept model, using IoT technology to automate the health care Sector. The project should submit with project proposal and it should consist with;

- a. Project description
- b. Problem statement
- c. Proposed solution
- d. Project execution and Methodology
- e. Conclusion

End of the Paper